

Hormones

Aesthetics
AWARDS
Winner 2025

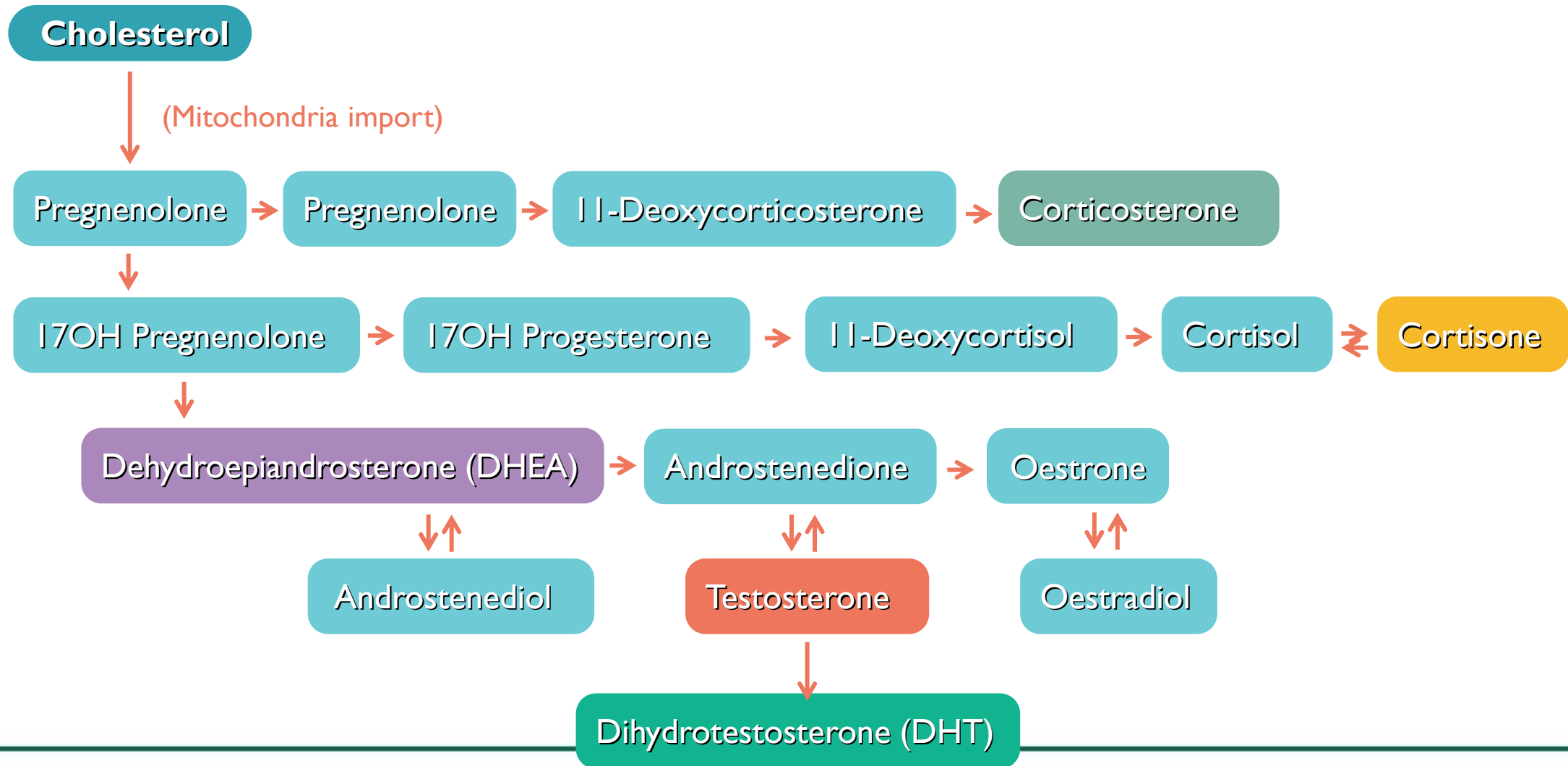


VTCT

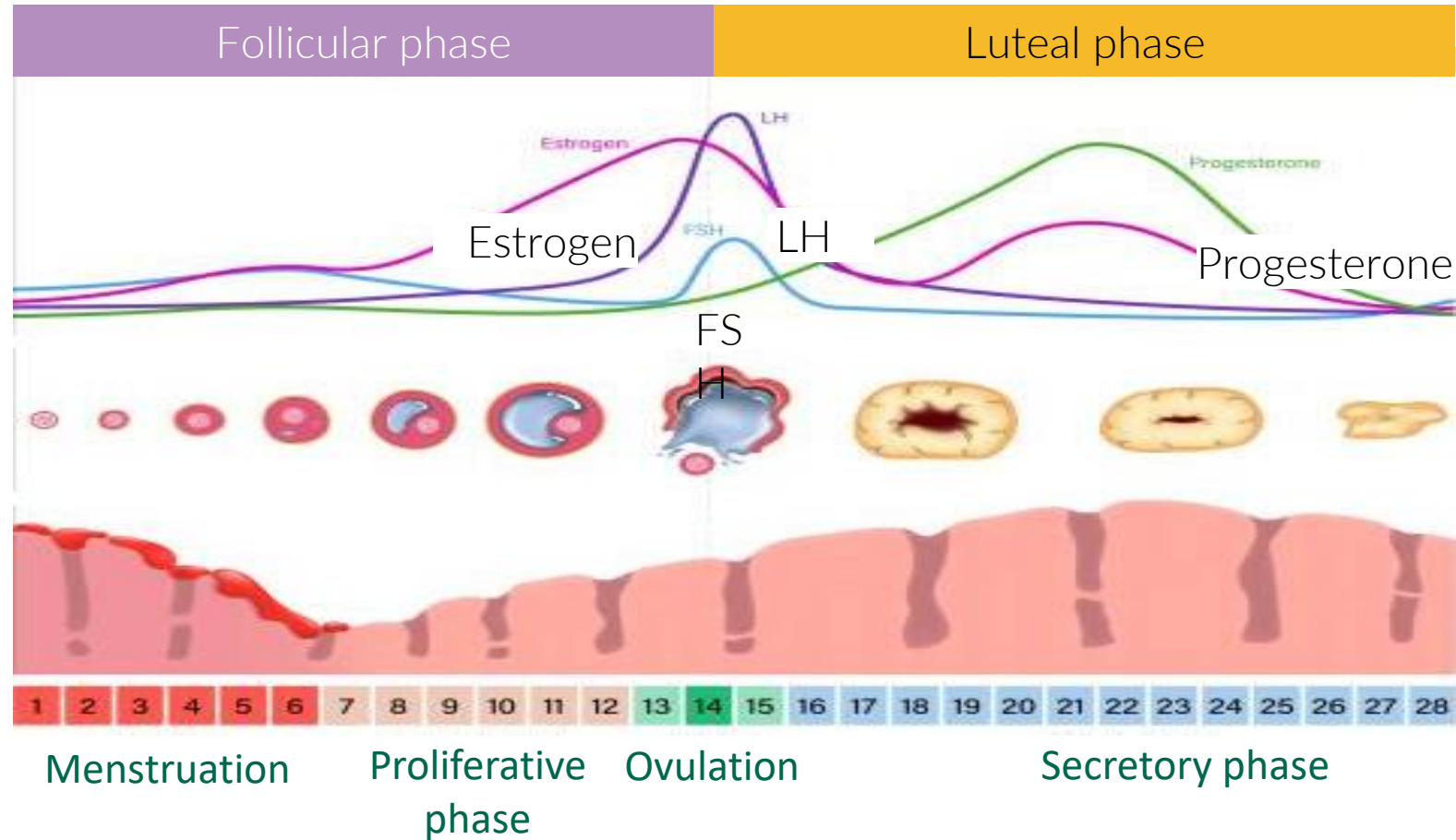


Cosmetic Courses
Teaching in medical aesthetics

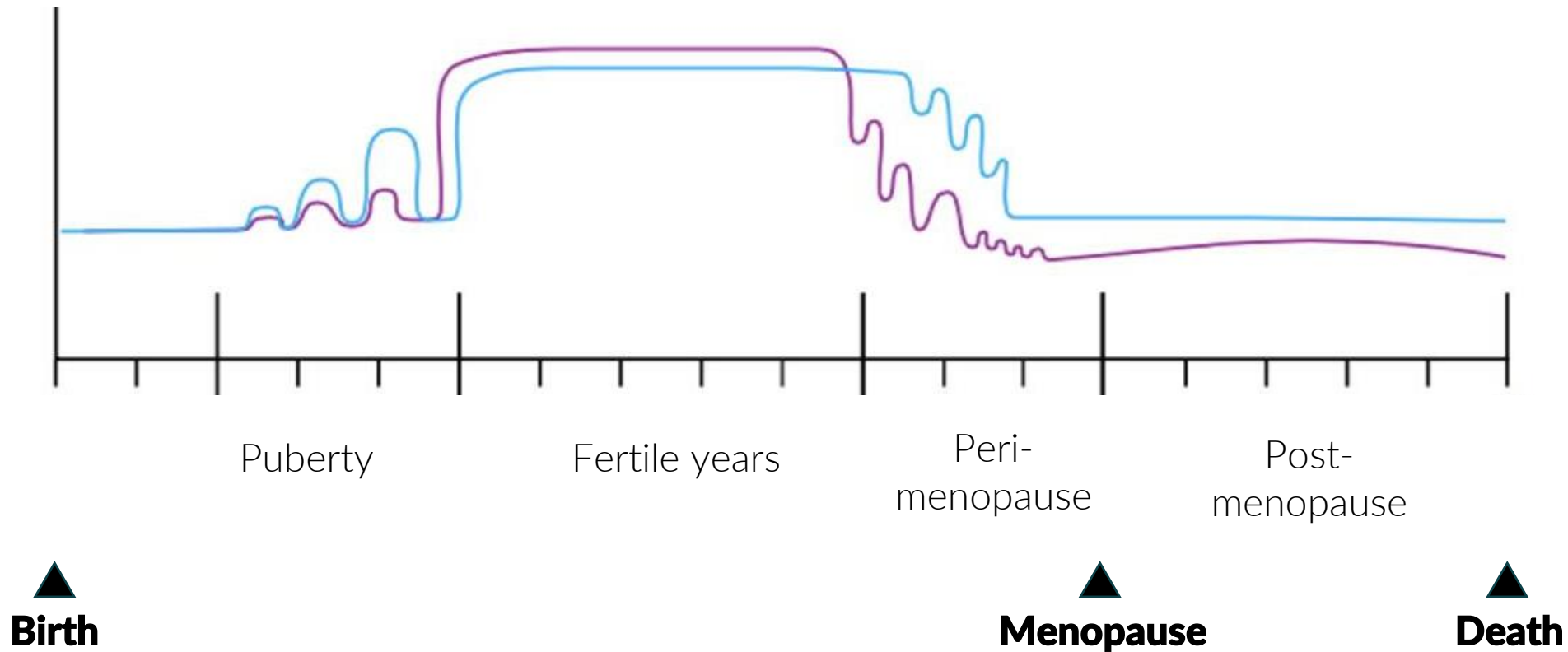
Hormone Cascade



Hormonal Fluctuations



Stages of Hormonal Changes



What is Menopause

- Menopause transition, begins several years before menopause. The ovaries don't produce oestrogens.
- It lasts up until menopause, the point when the ovaries stop releasing eggs.
- There are wide fluctuations in FSH (above 25 IU/L), oestrogens (above 200 pmol/L), and progesterone (below 5 nmol/L) causing the following symptoms:
 - Irregular periods
 - Breast tenderness
 - Fatigue
 - Weight gain
 - Aches and painful joints
 - Low libido
 - Insomnia
 - Mood swings, irritability
 - Anxiety
 - Frequent headaches

What is Menopause

- Early peri-menopause Women initially experience a change in menstrual cycle pattern when periods become infrequent and the cycles become slightly longer, e.g. 6-7 weeks apart. This phase is called the 'early peri-menopause'. The average age it occurs is 47 and it is when women may start experiencing menopausal symptoms.
- Late peri-menopause Subsequently, many women experience worsening of their menopausal symptoms. Menstrual cycles become less frequent, with periods a few months apart. This is called the 'late peri-menopause' and the average age it occurs is 49.

What is Menopause

- There is a gradual decline of hormone production over the years, this decline in hormone levels such as oestrogens, progesterone, and testosterone leads to symptoms.
- Menopause occurs when there hasn't been menstrual flow for a minimum of 12 months.
- The normal age to go through menopause ranges from 45 to 55, averaging 51 or 52.
- It is unpredictable how often women will experience symptoms and how severe they will be.

Menopause Symptoms

- Hot Flashes
- Night sweats
- Vaginal dryness
- Anxiety
- Mood swings
- Irritability
- Insomnia
- Depression
- Loss of sexual interest
- Hair growth on the face
- Painful intercourse
- Panic attacks
- Weird dreams
- Urinary tract infections
- Vaginal itching
- Lower back pain
- Bloating
- Weight gain

Main Oestrogen Functions

The woman's ovaries make most oestrogen hormones, although the adrenal glands and fat cells also make small amounts of the hormones.

- **Stimulates the production of choline acetyltransferase**
- Increases metabolic rate
- Improves insulin sensitivity
- **Regulates body temperature**
- Helps prevent muscle damage
- Helps maintain muscle
- **Improves sleep**
- Helps with some vascular actions.
- Decreases the accumulation of plaque on arteries
- Enhances magnesium uptake and utilization
- **Maintains the amount of collagen in the skin**
- Increases the water content of skin and is responsible for its thickness and softness

Main Oestrogen Functions

- Decreases LDL and prevents its oxidation
- Increases HDL by 10 to 15%
- Enhances the production of nerve growth factor
- Decreases lipoprotein(a)
- **Enhances energy**
- **Improves mood**
- Increases concentration
- Helps maintain memory
- Increases reasoning and new ideas
- **Maintains bone density**
- Helps with fine motor skills
- Helps prevent glaucoma
- **Increases sexual interest**
- Reduces homocysteine
- Protects against macular degeneration
- Decreases risk of colon cancer
- Helps prevent tooth loss
- Aids in the formation of neurotransmitters in the brain such as serotonin

Effects of Progesterone

- The main principal target organs of progesterone are the **uterus, the breasts, and the brain**
- Progesterone has an anti-estrogenic effect on the myometrial cells, decreasing their excitability, their sensitivity to oxytocin, and their spontaneous electrical activity while increasing their membrane potential.

Progesterone and its Functions

It is produced in the corpus luteum in the luteal phase of the menstrual cycle and by the placenta during pregnancy

Although it isn't used for other reasons apart from uterus protections, it is important to understand all the functions and make an informed decision about the treatment offered.

- Increases scalp hair
- Helps balance fluids in the cells
- Increases the beneficial effects of oestrogen
- Increases metabolic rate
- Natural diuretic
- Natural antidepressant
- Is anti-inflammatory
- Can reduce hot flushes
- Improves cholesterol levels

Progesterone and its Functions

- Stimulates the production of new bone
- Precursor of cortisone synthesis of adrenal cortex
- Enhances the action of thyroid hormones
- Improves libido
- Helps restore proper cell oxygen levels
- Induces conversion of E1 to the inactive E1S form
- Is neuroprotective, promoting myelination
- Studies have shown that progesterone does NOT induce oestrogen-stimulated breast cell proliferation.
- Enhancing fat breakdown and carbohydrates cravings.